

Polarisation (Cambridge (CIE) A Level Physics): Revision Note



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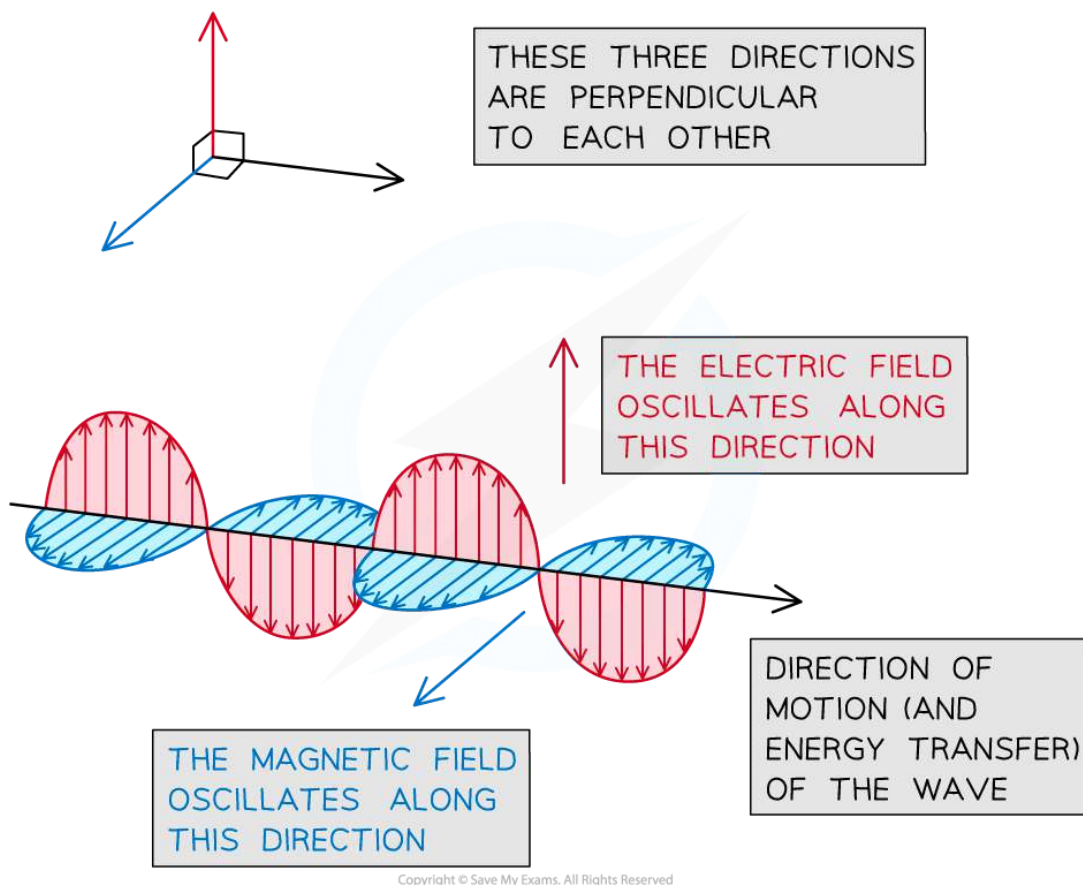
Reviewed by: Caroline Carroll

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Polarisation

Polarisation of transverse waves

- **Transverse waves** are waves that oscillate with their displacement perpendicular to their direction of travel
 - These oscillations can happen in **any plane** perpendicular to the propagation direction
- An **electromagnetic wave** is an example of a **transverse wave** that oscillates in **more than one plane**
 - A magnetic field oscillates on one plane
 - An electric field oscillates on a different plane
 - Both planes are at right angles to each other and to the direction of wave motion



An electromagnetic wave is generated by the combined oscillation of an electric and a magnetic field

- When transverse waves are **polarised**
 - vibrations are restricted to **one** plane of oscillation
 - vibrations are still **perpendicular** to the direction of propagation / energy transfer
- The difference between unpolarised and polarised waves are shown in the diagram below
 - The blue arrows represent the direction of the **plane of oscillation** of the transverse wave
- An **unpolarised** wave oscillates in **many planes** of oscillation
- A **vertically polarised** wave oscillates only in the **vertical plane**
- A **horizontally polarised** wave oscillates only in the **horizontal plane**

Different planes of polarisation

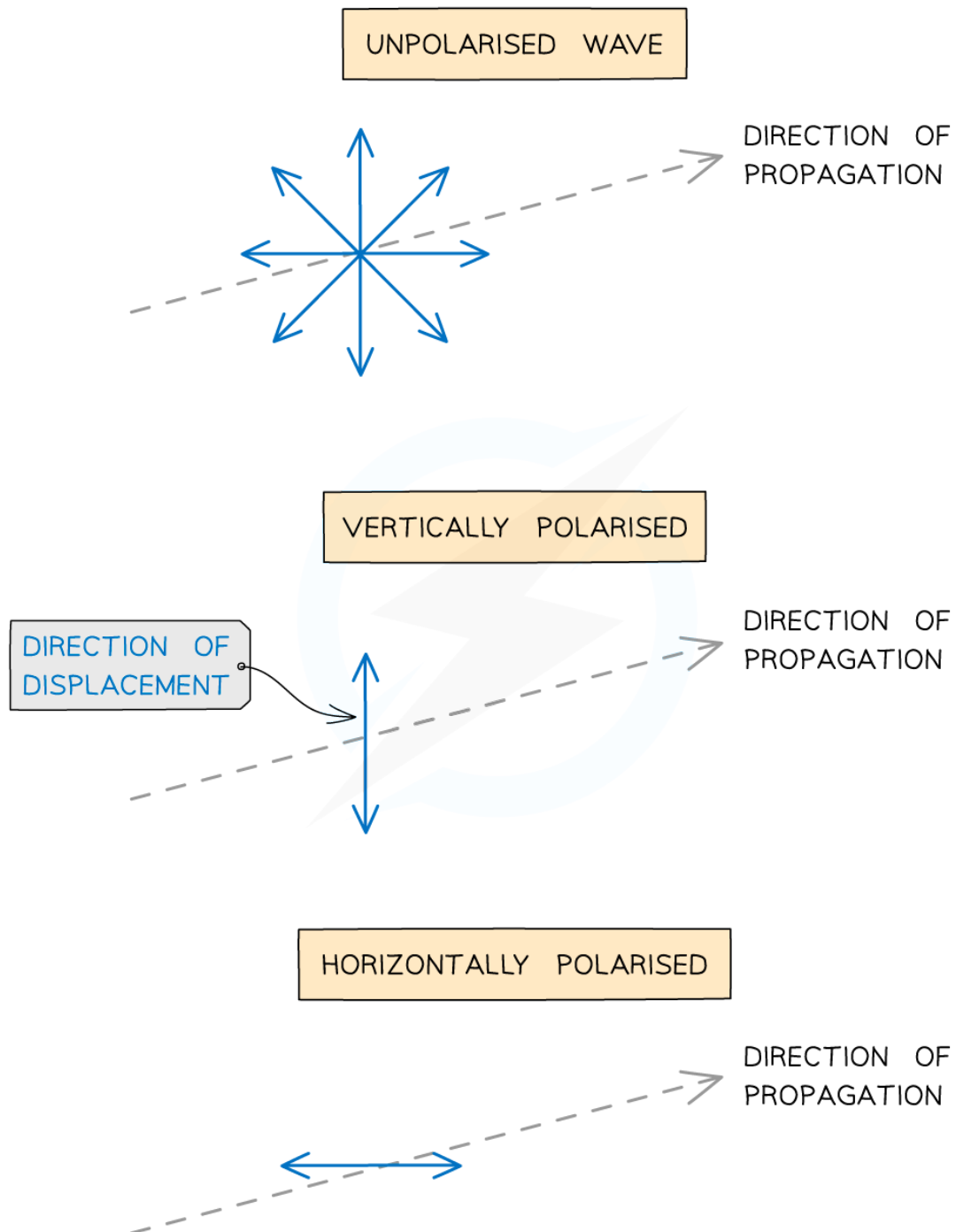


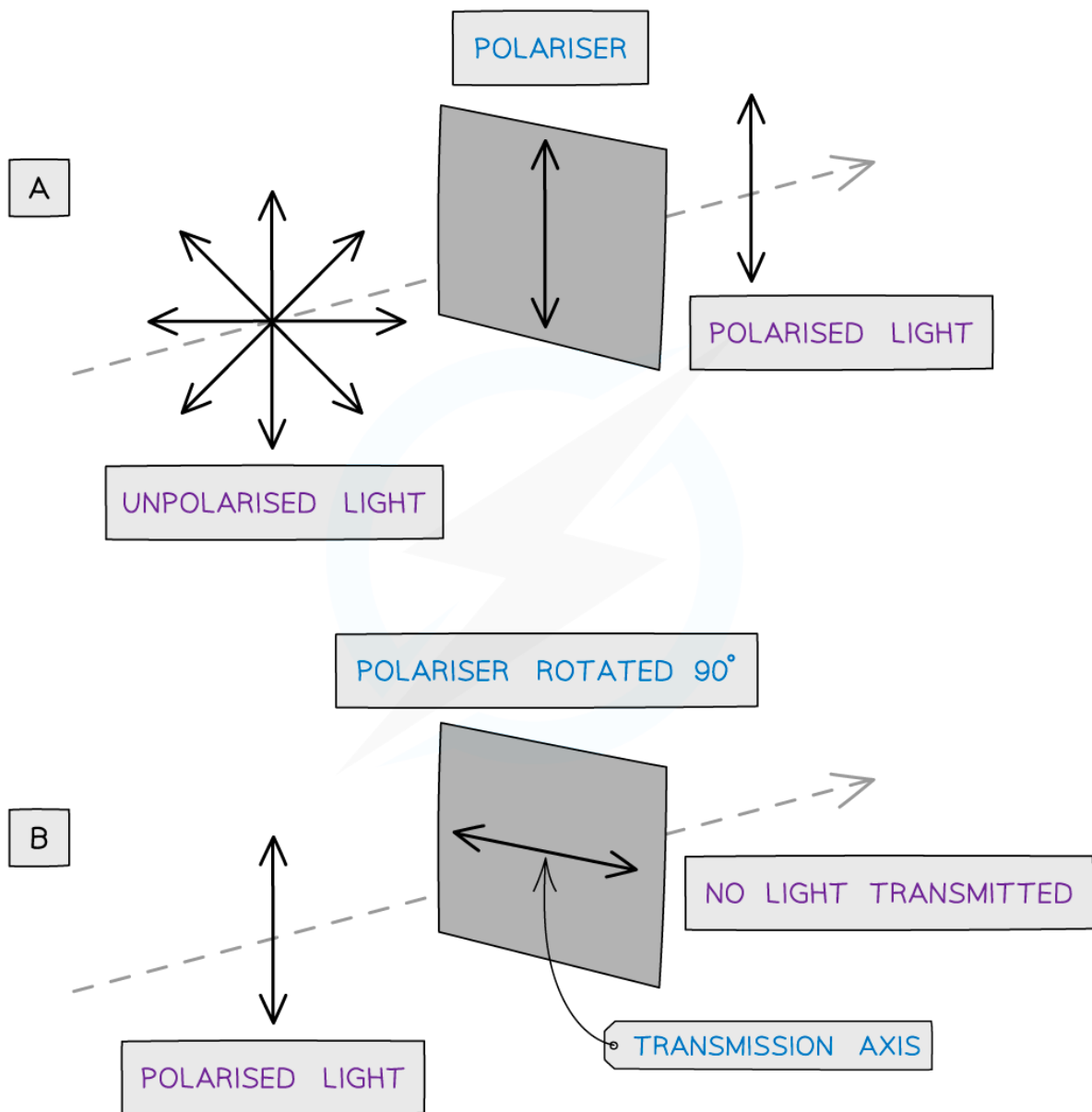
Diagram showing the displacement of unpolarised and polarised transverse waves

Polarising filters

- Waves can be polarised by passing through a **polariser** or **polarising filter**.
 - Polarisers only allow oscillations in a **certain plane** to be transmitted through the filter
- A **polariser** is an optical layer or lens that consists of **many parallel tiny slits**

- A polariser only allows waves vibrating parallel to the direction of these slits to pass through
- Waves that are not orientated in the correct plane are blocked by the filter and do not pass through

Action of a polarising filter



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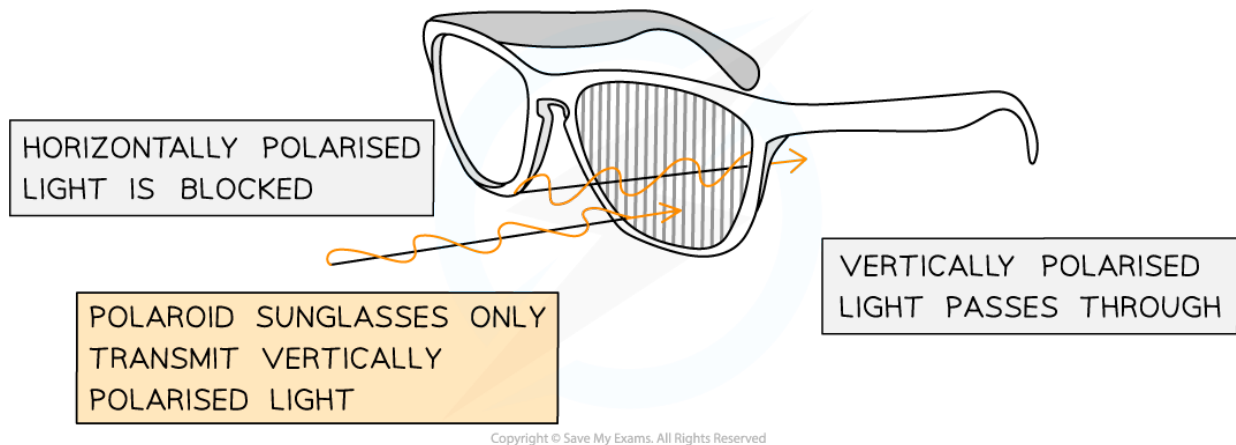
Diagram showing an unpolarised and polarised wave travelling through polarisers

- Only unpolarised waves can be polarised
 - This is shown in diagram A
 - When a polarised wave passes through a filter with a **transmission axis perpendicular** to the wave (diagram B), **none** of the wave will **pass through**

Polaroid sunglasses

- **Polaroid sunglasses** are an example of a polarisation
- Polaroid sunglasses contain lenses with polarising filters with transmission axes that are **vertically oriented**
 - This means the glasses do not allow any **horizontally polarised light** to pass through

Polaroid sunglasses



Polaroid sunglasses contain vertically oriented polarising filters which block out any horizontally polarised light



Worked Example

The following are statements about waves.

Which statement below describes a situation in which polarisation should happen?

- A.** Radio waves pass through a metal grid
- B.** Surface water waves are diffracted
- C.** Sound waves are reflected
- D.** Ultrasound waves pass through a metal grid

Answer: A

- Polarisation only occurs for transverse waves, therefore, **C** and **D** can be ruled out as sound and ultrasound are both longitudinal waves
- Waves are not polarised when diffracted, hence we can also rule out option **B**
- Radio waves are transverse waves - they can be polarised by a metal grid so only the waves that fit through the grid will be transmitted, therefore, **A** is correct

Malus's law

Intensity of polarised electromagnetic waves

- The intensity of electromagnetic waves is reduced as a result of polarisation
- If unpolarised electromagnetic waves of intensity I_0 pass through a polariser, the **intensity of the transmitted polarised waves falls by a half**

- $I = \frac{I_0}{2}$ is known as the **half rule**

Intensity of polarised and unpolarised light

The intensity of polarised light transmitted by a polariser is half the intensity of the unpolarised light incident on it

Intensity of analysed electromagnetic waves

- A second polarising filter placed after the first is known as an **analyser**

Intensity of waves when polariser and analyser have the same orientation

When the polariser and the analyser have the same orientation (i.e. parallel transmission axes), the intensity of analysed light is the same as the intensity of polarised light

Malus's law

- **Malus's law** states that if the analyser is rotated by an angle θ **with respect to the polariser**, the intensity of the waves transmitted by the analyser is given by the equation

$$I = I_0 \cos^2(\theta)$$

- Where
 - I is intensity of transmitted light in W m^{-2}
 - I_0 is maximum intensity (same units)

- If the polariser and the analyser have the **same orientation**, electromagnetic waves transmitted by the analyser have the **same intensity** as waves that are incident upon it, since $\cos(0) = 1$
 - If **vertically polarised** electromagnetic waves with intensity $\frac{I_0}{2}$ are incident on an analyser with a **vertical transmission axis**, **all** of the **waves** will be **transmitted** through the analyser
 - The **intensity** between the polariser and the analyser will **not decrease**
- If the analyser is rotated by **90°** with respect to the polariser ($\theta = 90^\circ$), the intensity of the light transmitted by the analyser will be **zero**, since $\cos(90^\circ) = 0$
 - If **vertically polarised light** is incident on an **analyser** with a **horizontal transmission axis**, none of the light will be transmitted through the analyser
 - In this instance, **all** the waves will be **absorbed**

Table of transmission depending on polariser orientation

Angle of transmission axis, θ / degrees	Direction of transmission axis	$\cos^2\theta$	Transmitted intensity / W m^{-2}	Maximum or minimum light intensity transmitted
0	Vertical	1	I_0	Max
180	Vertical	1	I_0	Max
90	Horizontal	0	0	Min
270	Horizontal	0	0	Min



Worked Example

Unpolarised light is incident on a polariser.

The light transmitted by the first polariser is then incident on a second polariser.

The polarising (or transmission) axis of the second polariser is 30° to that of the first. The intensity incident on the first polariser is I .

What is the intensity emerging from the second polariser?

A. $0.75 I$

B. $0.38 I$

C. $0.87 I$

D. $0.43 I$

Answer: B

Step 1: Apply the half rule:

- When light passes through the first polariser, half its intensity is lost

$$I_1 = \frac{1}{2}I$$

Step 2: Use Malus's law:

- The angle of the transmission axis is 30°
- In this case, I_0 in Malus's equation is I_1 here, because the light has already passed through the first polariser

$$I_2 = I_1 \cos^2(\theta) = \frac{1}{2}I \cos^2(30^\circ) = \frac{1}{2}I \times \frac{3}{4}$$

$$I_2 = \frac{3}{8}I = 0.375 I$$

- To 2 significant figures, this is option **B**



Examiner Tips and Tricks

Remember when using Malus's law to **square** the cosine of the angle ($\cos^2 \theta$)

Remember that the unpolarised light coming through will always halve in intensity when it becomes polarised through an polariser. Only **then** should you use Malus' law

to find the intensity of the light after it has passed through the analyser. Therefore, the I and I_0 in Malus' law are the intensities of light that are already polarised.